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Congestion-Aware Collective Algorithm Synthesis for *Heterogeneous* Multi-GPU Clusters

Animesh Saxena · University of Illinois Chicago
Ratul Sikder · University of Illinois Chicago
Balajee Vamanan · University of Illinois Chicago
Distributed ML Systems · Collective Communication

SYSTEMS FOR ML
DISTRIBUTED TRAINING
COLLECTIVE COMM.
NETWORK SIMULATION

Collective communication is a dominant bottleneck in distributed ML training. Existing libraries rely on fixed algorithm menus designed for homogeneous topologies. We present a simulation framework for evaluating topology-aware collective algorithm synthesis under realistic network conditions, extending an open-source distributed ML simulator with a congestion-aware analytical backend. We further propose extending this framework to multi-workload scenarios, exploiting idle network capacity through work-conserving scheduling.

01 · MOTIVATION

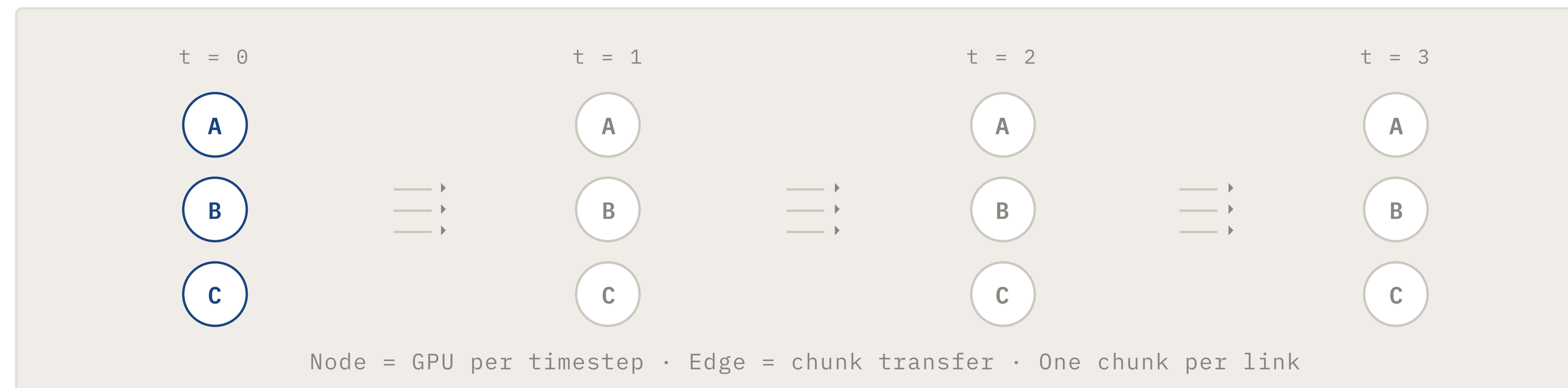
The Fixed Algorithm Problem

Collective libraries like NCCL rely on a fixed menu — Ring, Tree, Recursive Halving-Doubling — designed for **homogeneous, symmetric** topologies. Modern AI clusters use **heterogeneous, asymmetric interconnects**. Applying the wrong algorithm causes severe link oversubscription and underutilization.

02 · APPROACH

Synthesis via Time-Expanded Network

Given a topology and collective pattern, our framework synthesizes an optimized routing algorithm automatically. The **Time-Expanded Network (TEN)** replicates the physical topology across timesteps — a chunk moving GPU A→B at t=1 becomes edge (A,t=1)→(B,t=2).



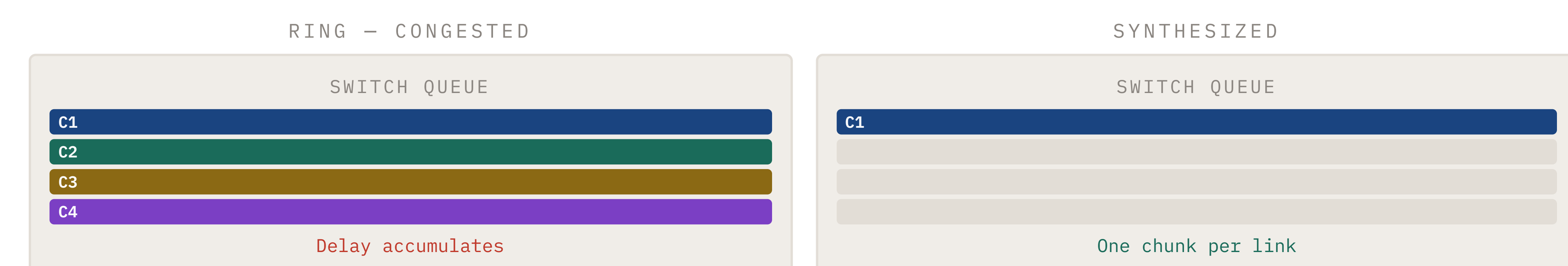
Greedy matching at each timestep assigns one chunk per link. Each link carries at most one chunk — **congestion-free by construction** for a single workload.

03 · CONTRIBUTION

Congestion-Aware Simulation Backend

Prior simulation infrastructure lacked congestion modeling, making fair evaluation against baselines impossible. We add **FIFO queuing at switches** using the α-β link cost model:

```
delay = α + β × chunk_size + queuing_delay
```

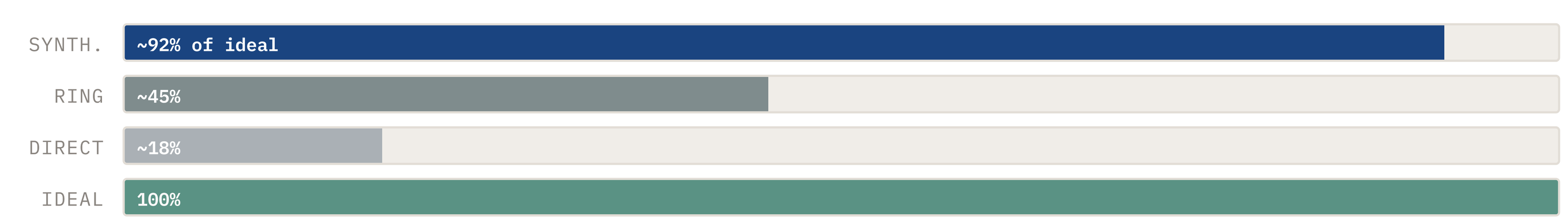


Key: PFC pause propagation in real RoCE fabrics affects only congested baselines — our results are therefore **conservative**, understating the true advantage of synthesis.

04 · RESULTS

Evaluation & Key Findings

ALL-REDUCE BANDWIDTH — NORMALIZED TO IDEAL

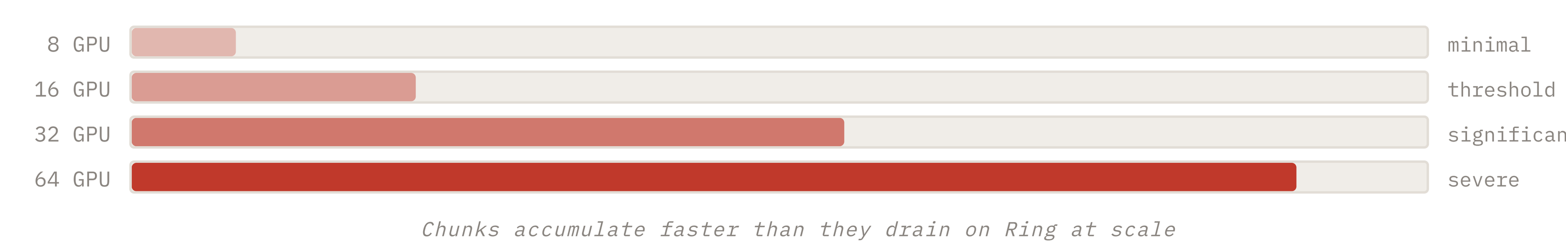


~5×
speedup over Ring at scale

90%+
of theoretical ideal BW

30%
idle capacity observed

RING QUEUE DEPTH VS CLUSTER SCALE



Chunks accumulate faster than they drain on Ring at scale

Synthesized algorithms maintain **near-zero queue depth** across all scales — congestion-free by construction. Baseline degradation validates the congestion model correctness.

05 · FUTURE DIRECTION

Handling Multiple Simultaneous Workloads

Real cloud clusters run multiple training jobs concurrently. A collective algorithm optimal in isolation becomes suboptimal when workloads contend for the same links. We observe **~30% idle links** per timestep — exploitable by concurrent jobs.

T=1	T=2	T=3	T=4	T=5	T=6	T=7
JOB A	JOB A	JOB B	JOB A	JOB B	JOB A	JOB B

Work-conserving: Job B fills idle slots left by Job A

- **Priority-based** link assignment with minimum bandwidth floors
- Joint synthesis constrained by per-job bandwidth reservations
- Target: concurrent makespan significantly below sequential
- No job starvation — bounded slowdown guaranteed

KEY TAKEAWAYS

I Fair Evaluation Requires Congestion Modeling

Without switch-level queuing, baseline algorithms appear artificially competitive. Our FIFO backend enables rigorous comparison — and because PFC cascades in real RoCE fabrics penalize only baselines, our results are conservative bounds.

II Topology-Aware Synthesis Scales Efficiently

Greedy TEN-based synthesis achieves 90%+ of theoretical ideal bandwidth in polynomial time $O(n^2)$, compared to NP-hard ILP approaches limited to tens of nodes. Heterogeneous and asymmetric topologies are fully supported.

III Idle Capacity Enables Multi-Job Scheduling

~30% of links idle per timestep in single-workload synthesis reveals significant slack. Work-conserving scheduling with minimum bandwidth guarantees is a practical path to higher cluster utilization without starvation.

Animesh Saxena · University of Illinois Chicago · PhD Student
Ratul Sikder · University of Illinois Chicago · PhD Student
Advisor: Balajee Vamanan · Department of Computer Science

asaxe26@uic.edu
linkedin.com/in/animeshsaxena
rsikd2@uic.edu
linkedin.com/in/ratulsikder121